TQQQ/SQQQ Systematic Trading Strategy - Logic Implementation

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1. System Overview

- Automated, end-of-day systematic strategy trading TQQQ (3+ long Nasdaq-100) and SQQQ (3+ short Nasdaq-100).
- Decision time: Market close (4:00 PM ET). System computes signals at ~3:59 PM and executes immediately.
- Signal source: $NDX daily close (Nasdaq-100 Index). In live trading, the 3:59 PM last price is used as the close proxy.
- Outputs: Target TQQQ % (0-100) and target SQQQ % (0-80).
- Constraint: TQQQ % + SQQQ % ≤ 100. Remainder is cash. TQQQ and SQQQ are never held simultaneously.
- Trade frequency: ~1-2 trades per week, ~8-10 per month.

2. Signal Timing Convention

All indicators are computed on the current trading day's close.

- In backtesting: Signal price = execution price = the official daily close.
- In live trading: At 3:59 PM ET, the system takes the current last-traded $NDX price as a close proxy, computes all indicators, determines the target allocation, and submits market orders in the final minute before the 4:00 PM bell. The close proxy and actual close will differ by a negligible amount (<0.05% typical).

3. Indicators and Calculations

All indicators are computed on $NDX daily closing prices (current day per Section 2).

3.1 Core Indicators

Indicator	Formula	Purpose
MA_20	20-day simple moving average of $NDX Close	Short-term trend
MA_250	250-day simple moving average of $NDX Close	Long-term trend

EXT_20	(Close – MA_20) / MA_20 × 100	Short-term over/under-extension (%)
EXT_250	(Close – MA_250) / MA_250 × 100	Long-term over/under-extension (%)
ROC_10	(Close – Close_10_days_ago) / Close_10_days_ago × 100	10-day rate of change (%)
ROC_ACCEL	ROC_10_today – ROC_10_5_days_ago	Momentum acceleration (positive = accelerating, negative = decelerating)
MA_250_Slope	(MA_250_today – MA_250_20_days_ago) / MA_250_20_days_ago × 100	Long-term trend health (%)

3.2 Bollinger Bands

Parameter	Value
Period	20 days
Standard Deviations	2.0
BB_Upper	MA_20 + (2 × 20-day StdDev)
BB_Lower	MA_20 – (2 × 20-day StdDev)
BB_Width	(BB_Upper – BB_Lower) / MA_20 × 100

3.3 Momentum Phase Detection

Condition	Definition
Upper Band Ride	Close > BB_Upper for ≥ 3 of last 5 trading days
Lower Band Ride	Close < BB_Lower for ≥ 3 of last 5 trading days
BB Expansion	BB_Width_today > BB_Width_20_day_avg × 1.2
Momentum Phase (Bullish)	Upper Band Ride AND BB Expansion
Momentum Phase (Bearish)	Lower Band Ride AND BB Expansion

4. Regime Classification

The first decision each day is the $NDX regime (A / B / C / D).

4.1 Regime Decision Tree (Authoritative)

Is Close > MA_250?
→ YES → BULL REGIME
  → Is Close > MA_20?
    → YES → Regime A: Strong Uptrend
    → NO → Regime B: Pullback in Uptrend

  → [Trend health - applied in Section 7.1, not here]

→ NO (Close ≤ MA_250) → BEAR REGIME
  → Is Close < MA_20?

→ YES → Regime C: Strong Downtrend
→ NO → Regime D: Bounce in Downtrend

→ [Trend health - applied in Section 7.1, not here]

Boundary rule: Close exactly equal to MA_250 → Bear regime. Close exactly equal to MA_20 → treated as below MA_20 (Regime B or C depending on MA_250 side). These conservative defaults err on the side of caution.

4.2 Regime Summary

Regime	Condition	Bias
A - Strong Uptrend	Close > MA_20 AND Close > MA_250	TQQQ heavy
B - Pullback in Uptrend	Close ≤ MA_20 AND Close > MA_250	TQQQ moderate (mean reversion)
C - Strong Downtrend	Close < MA_20 AND Close ≤ MA_250	SQQQ heavy
D - Bounce in Downtrend	Close > MA_20 AND Close ≤ MA_250	SQQQ moderate (mean reversion short)

Clarification: Each condition checks Close against each MA independently. The relationship between MA_20 and MA_250 themselves is irrelevant to regime classification. For example, in an early recovery, Close can be above both MAs while MA_20 is still below MA_250 - this is Regime A. The Trend Health modifier (Section 7.1) handles conviction adjustment when the MA structure is weak.

5. Sub-Strategy Rules

5.1 Rule Selection Logic

Rules are priority-ranked within their group. Multiple rules can have their conditions met simultaneously. On each side (long/short), evaluate rules in the eval order shown (top to bottom). The first rule whose trigger conditions are fully met is selected.

- Only one long rule and one short rule can be active at a time.
- Only one side (long or short) holds a non-zero allocation (see Section 6.1).

5.2 Long Side (TQQQ) - Rule Definitions

Eval Order: 1
Rule: L4: Oversold Snap-Back
Regime: B
Trigger: Close < BB_Lower
Logic: Extreme short-term oversold within intact long-term uptrend. Betting on a bounce.
Exit: Revert to L3/L2 when Close ≥ BB_Lower. Exit if Close ≤ MA_250 (regime shift).

Eval Order: 2

Rule: L3: Deep Pullback
Regime: B
Trigger: EXT_20 ≤ −3%
Logic: Deeper pullback. Price moving toward MA_250. Higher risk, smaller position.
Exit: Upgrade to L2 on recovery. Exit if Close ≤ MA_250 (regime shift).

Eval Order: 3
Rule: L2: Shallow Pullback
Regime: B
Trigger: EXT_20 > −3% (and Close ≤ MA_20, implied by Regime B)
Logic: Healthy pullback still close to MA_20.
Exit: Revert to L1 when Close > MA_20. Downgrade to L3 if pullback deepens.

Eval Order: 4
Rule: L1: Trend Following
Regime: A
Trigger: Regime A (always active when Regime A)
Logic: Core trend-following position. Ride the uptrend.
Exit: Downgrade when regime changes.

Why this order: L4 is the most extreme/specific condition and is checked first. L3 and L2 partition the remaining Regime B space by EXT_20 with no gap. L1 is the Regime A default.

5.3 Long Side - Position Sizing

Each table uses top-to-bottom, first-match evaluation. Conditions within each table are non-overlapping.

L1 (Trend Following - Regime A)

Eval Order	Condition	Base Size
1	ROC_ACCEL > 0	80% TQQQ
2	ROC_ACCEL ≤ 0 AND EXT_20 > 5%	40% TQQQ (trim - extended and decelerating)
3	ROC_ACCEL ≤ 0 AND EXT_20 ≤ 5%	60% TQQQ

L2 (Shallow Pullback - Regime B, EXT_20 > −3%)

Eval Order	Condition	Base Size
1	EXT_250 > 5%	50% TQQQ
2	EXT_250 ≤ 5%	40% TQQQ

L3 (Deep Pullback - Regime B, EXT_20 ≤ −3%)

Eval Order	Condition	Base Size
1	EXT_20 > −6%	25% TQQQ
2	EXT_20 ≤ −6%	10% TQQQ

L4 (Oversold Snap-Back - Regime B, Close < BB_Lower)

Eval Order	Condition	Base Size
1	Days below BB_Lower = 1–2	20% TQQQ
2	Days below BB_Lower ≥ 3	10% TQQQ (trend may be breaking)

5.4 Short Side (SQQQ) - Rule Definitions

Eval Order: 1
Rule: S2: Breakdown Short
Regime: C
Trigger: Close broke below MA_250 within last 10 trading days AND ROC_ACCEL < 0 (see S2 Break Definition below)
Logic: Fresh breakdown with accelerating downside momentum. Most aggressive short.
Exit: Reverts to S1 after 10 days or if conditions no longer met. Exit if Close reclaims MA_250.

Eval Order: 2
Rule: S1: Trend Following Short
Regime: C
Trigger: Regime C (always active when Regime C)
Logic: Core trend-following short. Default when S2 conditions aren't met.
Exit: Downgrade when regime changes.

Eval Order: 3
Rule: S3: Overbought Mean Reversion Short
Regime: D
Trigger: Regime D (always active when Regime D)
Logic: Relief rally expected to fail in bear regime.
Exit: Revert to S1 when Close < MA_20. Exit if Close > MA_250 (regime shift).

Note: Max SQQQ allocation is 80%, not 100%. Asymmetric by design - markets have a long-term upward bias.

Why this order: S2 is the most specific condition in Regime C (fresh breakdown + accelerating downside). If S2's conditions aren't met, S1 serves as the Regime C fallback. S3 applies exclusively in Regime D.

S2 Break Definition

A "break" is the first close ≤ MA_250 following a close that was > MA_250. The day counter starts at 0 on the break day and increments each trading day. If Close crosses back above MA_250 at any point during the 10-day window, S2 is invalidated - the whipsaw filter takes over (exit to cash, wait for 2 consecutive closes). If a new break occurs after the filter clears, it is treated as a new S2 event with the counter reset to 0.

The 10-day window counts from the break date, including days spent in the whipsaw filter. Since the filter requires 2 days of cash, S2's effective trading window is approximately 8 days. This is intentional - if the breakdown signal has not played out within 10 calendar trading days of the actual break, it is no longer "fresh."

5.5 Short Side - Position Sizing

S1 (Trend Following Short - Regime C, default)

Eval Order	Condition	Base Size
1	ROC_ACCEL < 0	60% SQQQ
2	ROC_ACCEL ≤ 0 AND EXT_20 < −5%	25% SQQQ (trim - stretched and slowing)
3	ROC_ACCEL ≤ 0 AND EXT_20 ≤ −5%	40% SQQQ

S2 (Breakdown Short - Regime C, fresh breakdown)

Eval Order	Condition	Base Size
1	Days since break ≤ 5 AND BB Expansion active	70% SQQQ
2	Days since break ≤ 5 AND BB Expansion NOT active	50% SQQQ
3	Days since break 6–10 AND BB Expansion active	60% SQQQ (sustained selling pressure)
4	Days since break 6–10 AND BB Expansion NOT active	50% SQQQ

S3 (Overbought Mean Reversion Short - Regime D)

Row 1 is an override - if Close > BB_Upper, it takes precedence over all other S3 conditions regardless of EXT_20 or ROC_ACCEL.

Eval Order	Condition	Base Size
1	Close > BB_Upper (override)	50% SQQQ (extreme overbought in downtrend)
2	EXT_20 ≥ 3% AND ROC_ACCEL ≤ 0	40% SQQQ (extended bounce losing steam)
3	EXT_20 ≥ 3% AND ROC_ACCEL > 0	20% SQQQ (strong bounce - cautious)
4	EXT_20 < 3% AND ROC_ACCEL ≤ 0	30% SQQQ (modest bounce fading)
5	EXT_20 < 3% AND ROC_ACCEL > 0	15% SQQQ (bounce gaining - very cautious)

6. Position Aggregation and Side Selection

6.1 Side Selection (TQQQ vs SQQQ vs Cash)

Is Regime A or B?

→ YES → Long side active. SQQQ = 0%.

Select the first matching long rule (Section 5.2).
TQQQ % = rule output (after adjustments in Section 7).
Cash % = 100% − TQQQ %.

→ NO (Regime C or D)
→ Short side active. TQQQ = 0%.
Select the first matching short rule (Section 5.4).
SQQQ % = rule output (after adjustments in Section 7).
Cash % = 100% − SQQQ %.

6.2 Regime Transition Handling

Transition: A → B
Type: Same side (bull)
Immediate Action: Reduce TQQQ per L2/L3/L4 sizing. No SQQQ.
Then: Continue in Regime B.

Transition: B → A
Type: Same side (bull)
Immediate Action: Increase TQQQ per L1 sizing.
Then: Continue in Regime A.

Transition: C → D
Type: Same side (bear)
Immediate Action: Reduce SQQQ per S3 sizing. No TQQQ.
Then: Continue in Regime D.

Transition: D → C
Type: Same side (bear)
Immediate Action: Increase SQQQ per S1/S2 sizing.
Then: Continue in Regime C.

Transition: B → C
Type: MA_250 cross
Immediate Action: Exit all TQQQ → 100% cash.
Then: Enter whipsaw filter (Section 6.3).

Transition: D → A
Type: MA_250 cross
Immediate Action: Exit all SQQQ → 100% cash.
Then: Enter whipsaw filter (Section 6.3).

Transition: A → C

Type: MA_250 cross (crash)
Immediate Action: Exit all TQQQ → 100% cash.
Then: Enter whipsaw filter.

Transition: A → D
Type: MA_250 cross
Immediate Action: Exit all TQQQ → 100% cash.
Then: Enter whipsaw filter.

Transition: C → A
Type: MA_250 cross (rip)
Immediate Action: Exit all SQQQ → 100% cash.
Then: Enter whipsaw filter.

Transition: C → B
Type: MA_250 cross
Immediate Action: Exit all SQQQ → 100% cash.
Then: Enter whipsaw filter.

Key rule: Any transition that crosses MA_250 triggers an immediate exit to cash and the whipsaw filter. Transitions within the same side of MA_250 (A–B, C–D) execute normally - no filter, no cash buffer.

6.3 Whipsaw Filter (MA_250 Cross)

- On the day Close crosses MA_250, immediately exit all positions to 100% cash. Do not hold the prior position.
- Require 2 consecutive confirmed closes on the new side of MA_250 before committing capital to the new direction.
- During the waiting period (days 1–2 after cross), the system holds 100% cash. No positions on either side. Post-rule adjustments (Section 7) and the intraday stop-loss (Section 8.6) do not apply - there is nothing to adjust.
- On the 3rd day (after 2 consecutive closes confirm the new side), the system evaluates the new regime's rules and enters the indicated position.
- If during the waiting period Close crosses back to the original side, the filter resets. The system remains in cash until 2 consecutive closes confirm a direction.

7. Post-Rule Adjustments

After the active rule produces a base position size (Section 5), two adjustments are applied sequentially before risk guardrails. These adjustments do not apply during the whipsaw filter cash period (position is zero).

7.1 Trend Health Modifier (MA_250_Slope)

Regime	Condition	Multiplier

Bull (A or B)	MA_250_Slope > 0 (uptrend intact)	1.0×
Bull (A or B)	MA_250_Slope ≤ 0 (uptrend weakening)	0.75×
Bear (C or D)	MA_250_Slope < 0 (downtrend intact)	1.0×
Bear (C or D)	MA_250_Slope ≥ 0 (downtrend weakening)	0.75×

7.2 Momentum Phase Overlay

Momentum Phase	Applies To	Multiplier	Cap
Bullish Momentum Phase	Long side (TQQQ)	1.25×	100%
Bearish Momentum Phase	Short side (SQQQ)	1.25×	80%
No Momentum Phase	Either side	1.0×	-

Final Position = min(Base Size × Trend Health Multiplier × Momentum Multiplier, Cap)

7.3 Adjustment Computation (Step-by-Step)

- Example 1 - Full acceleration in healthy uptrend with momentum: Rule L1, ROC_ACCEL > 0 → Base = 80%. Trend Health = 1.0× → 80%. Momentum = 1.25× ≤ 100%. Cap = 100% TQQQ.
- Example 2 - Decelerating in weakening uptrend, no momentum: Rule L1, ROC_ACCEL ≤ 0, EXT_20 ≤ 5% → Base = 60%. Trend Health = 0.75× → 45%. Momentum = 1.0× → 45%. Cap = 45% TQQQ.
- Example 3 - Fresh breakdown short with expansion: Rule S2, Days ≤ 5 AND BB Expansion → Base = 70%. Trend Health = 1.0× → 70%. Momentum = 1.25× → 87.5%. Cap = 80% SQQQ.

8. Risk Management

8.1 Maximum Drawdown Circuit Breaker

Peak equity is defined as the highest end-of-day portfolio value since system inception or since the last manual restart after a halt.

Drawdown from Peak Equity	Action
−30%	Alert only. System continues. (Expected per backtest.)
−40%	Reduce all positions to 50% of computed signal. Alert client.
−50%	Halt system. Go 100% cash. Require manual restart by client.

8.2 Daily Loss Limit

- Condition: Single-day portfolio loss > 15% → Reduce-only mode next trading day: system can reduce or exit positions but cannot increase or initiate new positions. Existing positions held at current size or lower.
- Condition: Two consecutive days with > 10% loss each → Reduce all positions to 50% of computed signal for the next 5 trading days.

Daily loss is measured using end-of-day portfolio values (prior close to current close). Intraday drawdowns that recover by close do not trigger this rule. If intraday risk protection is needed, enable the intraday stop-loss module (Section 8.6).

8.3 Position Size Guardrails

Rule	Constraint
Max TQQQ allocation	100%
Max SQQQ allocation	80%
Min position change to execute	5% of portfolio (avoid micro-trades from noise)
No simultaneous TQQQ + SQQQ	TQQQ % × SQQQ % must always = 0

Exception: The 5% minimum trade threshold applies to signal-driven rebalancing only. Risk-driven actions - circuit breaker reductions, reduce-only exits, halt-to-cash, whipsaw exits, and intraday stop-loss triggers (Section 8.6) - execute regardless of delta size.

8.4 Risk Rule Precedence

- If drawdown circuit breaker says "reduce to 50%" AND daily loss limit says "reduce to 50% for 5 days" → apply 50% reduction with the longer duration (5 trading days).
- If drawdown circuit breaker says "halt" AND any other rule says "reduce" - halt wins. Go to 100% cash.
- Reduce-only mode (Section 8.2) stacks with drawdown reduction: if both trigger, apply 50% reduction AND enforce reduce-only (no increases even within the 50% budget).
- Intraday stop-loss (Section 8.6) overrides all other rules when triggered - it forces an immediate exit regardless of what EOD signals say.

8.5 Execution Risk

- Order type: Market order (MOC preferred if broker supports).
- Execution window: 3:59-4:00 PM ET (final minute).
- Retry logic: If order fails, retry every 15 seconds until 4:00 PM.
- Fallback (order): If unable to execute by close, execute at next open via market order.
- Data failure: If $NDX data is unavailable at 3:59 PM, hold current positions (no trades). Log failure and alert client.

8.6 Intraday Stop-Loss

This is an optional add-on to be backtested, that monitors positions during market hours and exits if losses exceed a threshold. It operates independently of the EOD system and can be enabled or disabled without affecting core strategy logic.

- Parameter: Stop-Loss Trigger. Default: −10% intraday P&L. Description: If the portfolio's intraday unrealized loss (from prior close) exceeds this threshold at any point during market hours.
- Parameter: Action. Default: Exit 100% to cash. Description: Immediately liquidate all TQQQ or SQQQ via market order.
- Parameter: Cooldown. Default: Remainder of current trading day. Description: No re-entry until the next EOD signal. The EOD system runs normally at close and may re-enter.
- Parameter: Monitoring Frequency. Default: Every 5 minutes, 9:30 AM–3:58 PM ET. Description: System polls portfolio P&L at this interval.

Interaction with EOD system

- If the intraday stop fires at, say, 1:00 PM, the system exits to cash.
- At 3:59 PM, the EOD system runs its normal daily flow. It sees the current position as 0% (cash). It computes the target allocation based on today's close. If the signal says to re-enter, it does. If the signal says cash, it stays in cash.
- The intraday stop does not alter regime classification, rule selection, or any indicators. It is purely a position-level emergency exit.
- If the intraday stop fires, it is logged and an alert is sent to the client.

When disabled: The system relies entirely on EOD risk management (Sections 8.1–8.4). No intraday monitoring occurs.

9. Complete Daily Decision Flow

```
[~3:59 PM ET - System Activates]
 |
 v
[Fetch $NDX current price as close proxy + prior data]
 |
 |-- Data unavailable? -> Hold current positions, alert client, STOP.
 |
 v
[Compute indicators: MA_20, MA_250, EXT_20, EXT_250,
 ROC_10, ROC_ACCEL, BB_Upper, BB_Lower, BB_Width, MA_250_Slope]
 |
 v
[Determine Regime: A / B / C / D]
 |
 v
[Check: currently in whipsaw filter waiting period?]
 |-- YES -> Hold 100% cash. STOP.
 |
 v
```

```
[Check: did regime cross MA_250 since last decision?]
  |-- YES -> Exit all positions to cash.
  |        Enter whipsaw filter. STOP.
  |
  v
[Check risk circuit breakers (drawdown + daily loss)]
  |-- Halt triggered? -> Go 100% cash. Alert client. STOP.
  |
  v
+-------------------- Regime Split ---------------------+
| Regime A or B:                Regime C or D:        |
| Evaluate L4 -> L3 -> L2 -> L1   Evaluate S2 -> S1 / S3|
| First match = active rule       First match = active  |
| Get base TQQQ %                 Get base SQQQ %       |
+-------------------------------------------------------+
  |
  v
[Apply Trend Health modifier (Section 7.1)]
  |
  v
[Apply Momentum Phase multiplier (Section 7.2)]
  |
  v
[Apply risk reductions if active (50% of signal, reduce-only)]
  |
  v
[Apply caps: TQQQ ≤ 100%, SQQQ ≤ 80%]
  |
  v
[Set opposite side to 0%. Compute cash %.]
  |
  v
[Compare target vs current holdings]
  |-- Signal-driven delta < 5% of portfolio -> No trade. Hold current.
  |   (Risk-driven actions exempt from this threshold.)
  |
  v
[Compute shares to buy/sell]
  |
  v
[Execute orders (final minute before close)]
  |
  v
```

```
[Log trade + send alert to client]
 |
 v
[Done]

If Intraday Stop-Loss is enabled (Section 8.6):
[9:30 AM–3:58 PM ET - Monitoring Loop]
 |
 Every 5 minutes:
 |-- Compute intraday P&L from prior close
 |-- P&L < −10% -> Liquidate all positions to cash.
 |            Log + alert. Enter cooldown.
 |            EOD system still runs at close.
 `-- Otherwise -> Continue monitoring.
```

10. Parameters Summary (Optimization Candidates)

All parameters below will be tested in Phase 3 (Backtesting & Optimization). Current values are logical starting points.

Parameter	Starting Value	Optimization Range	Sensitivity
Short MA Period	20	10–50	High
Long MA Period	250	1.0×300	Medium
ROC Lookback	10 days	5–20	Medium
ROC Acceleration Lookback	5 days	3–10	Medium
BB Period	20	14–30	Medium
BB Std Dev	2.0	1.5–2.5	Low
BB Momentum: days above/below band	3 of 5	2/5 to 4/5	High
BB Expansion Threshold	1.2× avg width	1.1–1.5	Medium
Whipsaw Filter Days	2 consecutive closes	1–3	High
EXT_20 threshold (L2/L3 boundary)	−3%	−2% to −5%	High
EXT_20 threshold (L3 extreme)	−6%	−5% to −8%	Medium
EXT_20 threshold (S3 boundary)	3%	2%–5%	High
EXT_20 trim threshold (L1/S1)	−5%	−3% to −7%	High
Trend Health Multiplier	0.75×	0.6–0.9	Medium
Momentum Multiplier	1.25×	1.1–1.5	Medium

Min Trade Threshold	5% delta	3–10%	Low
Max SQQQ Cap	80%	60–100%	Medium
Circuit Breaker: Reduce at	−40% drawdown	−30% to −50%	Low
Circuit Breaker: Halt at	−50% drawdown	−40% to −60%	Low
Intraday Stop-Loss (optional)	−10% intraday	−7% to −15%	Medium
Intraday Monitoring Frequency (optional)	5 min	1–15 min	Low

11. Assumptions

11.1 Assumptions Made

- The system is always invested or in cash. No margin or leverage beyond what TQQQ/SQQQ inherently provide.
- All indicators use the current day's close. In live trading, a 3:59 PM close proxy is used; in backtesting, the official close is used for both signal and execution.
- Position sizes are % of total equity (compounding - as equity grows, positions grow proportionally).
- Short-side cap at 80% reflects long-term upward equity bias. Adjustable per client preference.
- TQQQ and SQQQ are never held simultaneously. The system is always directional or in cash.
- 2-day whipsaw filter on MA_250 crosses adds safety but will occasionally cause missed entries. Tradeoff validated in backtesting.
- Boundary values (Close = MA_250, Close = MA_20) are treated as the "weaker" side (bear for MA_250, pullback for MA_20) to err on the side of caution.
- The intraday stop-loss module is optional and independent of core EOD logic.
- On first run, the system assumes 100% cash, peak equity = starting account value, no whipsaw filter active, no active risk cooldowns. Regime is determined and positions entered normally on the first trading day.